Housing Lending, Overseas Migration and Wage Growth Across Australian States

Data Exploration Project

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FIT5147 Data Exploration and Visualisation
Semester 1, 2026
Monash University
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Applied class: Applied 01

29 March 2026

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1 Introduction

This report, *Housing Lending, Overseas Migration and Wage Growth Across Australian States*, examines housing affordability in Australia, a major social and economic issue. Public debate often links housing pressure to migration, limited supply, and weak wage growth. For example, SBS News (2026) discussed whether high migration is truly supported by evidence as a cause of the housing crisis. This makes the topic important for households, workers, students, and policy makers.

The report explores how housing lending, overseas migration, and wage growth have changed across Australian states over time. Using Australian Bureau of Statistics data, it examines owner-occupier lending, investor lending, net overseas migration, and the Wage Price Index. The aim is to assess whether common public claims about migration, wages, and housing stress are reflected in the data, and whether these patterns differ by state.

I am personally interested in this topic because, as an international student from Taiwan living in Melbourne, I have become more aware of housing affordability pressures. Seeing how difficult high housing costs can make financial security for students and workers motivated me to examine this issue using data rather than relying only on public opinion or media debate.

This report addresses the following questions:

1. Does the growth of housing lending align with wage growth across Australian states, and how has the lending structure changed over time?
2. How have the origins of overseas migrants and their state settlement patterns changed across Australian states over time, and what are their links with wage growth and housing lending?
3. Where are the largest gaps between investor lending and migration demand across Australian states, and how have these gaps changed over time?

2 Data Wrangling and Checking

This project uses four open datasets from the Australian Bureau of Statistics (ABS): Net Overseas Migration by Country of Birth, Owner-Occupier Lending, Investor Lending, and the Wage Price Index. The direct download links are provided below:

- Net Overseas Migration by Country of Birth
- Owner-Occupier Lending
- Investor Lending
- Wage Price Index

Before checking and cleaning the data, I first converted the ABS Excel workbooks into tidy datasets that were easier to compare and analyse in R. Although the files came from similar ABS sources, their internal structures were quite different, so they could not be used directly.

For the overseas migration data, the workbook contained nine separate tables, including one national table and eight state or territory tables. I created a lookup table of sheet names, states, and row ranges, then used a function to read each sheet in the same way. I used the first imported row as the column names, renamed the first two columns as `sacc_code` and `country_of_birth`, removed note labels from the year columns, and reshaped the data from wide to long format using `pivot_longer()`. I then combined all nine sheets into one dataset called `overseas_migration`.

For the housing finance data, I used the `Data1` sheet from both the owner-occupier and investor lending workbooks. I read the first ten metadata rows separately from the time series data, created temporary series column names, and extracted variables such as state, value type, unit, and series type from the ABS series descriptions. I then reshaped the data into long format, created year, month, and financial year variables, and combined the two sources into one dataset called `housing_finance`, with a `loan_type` variable to show whether the series was owner-occupier or investor lending.

For the Wage Price Index data, I also used the `Data1` sheet and followed a similar process. I read the metadata rows separately, created temporary series column names, extracted variables such as state and wage measure, and reshaped the data into long format. I then created year, month, and financial year variables to produce a final tidy dataset called `wage_price_index`.

Overall, this step turned the original Excel workbooks into consistent long-format datasets that could be compared across states and time. It also made the later checking, filtering, and analysis much easier.

2.1 Creating a Common Financial Year Structure

As shown in Table 1, Housing finance and Wage price index are quarterly datasets recorded using calendar year and month, while Overseas migration is recorded by financial year. To make the three datasets easier to compare, I created a financial year variable for the quarterly datasets based on the quarter-ending months of September, December, March, and June. The datasets also have different time ranges, since Housing finance and Wage price index start earlier than Overseas migration. Therefore, I restricted all three datasets to the shared financial-year period from 2004-05 to 2024-25 before carrying out the analysis.

Table 1: Summary of time structure across the datasets

Dataset	Frequency	Time period
Overseas migration	Annual	2004-05 to 2024-25
Housing finance	Quarterly	2002-03 to 2025-26
Wage price index	Quarterly	1997-98 to 2025-26

2.2 Selecting variables for the housing finance and wage price index data

Table 2 shows that the housing finance master dataset includes both owner-occupier and investor lending series, as well as both Value and Number measures. The only later-starting series is Investor Number, which is available from 2019-20 onward, suggesting that these missing values are structural rather than accidental. However, the main analysis in this report uses only the Original Value series. These were the series used to calculate housing lending growth, investor shares, and investor-migration gaps. Seasonally adjusted and Trend series were not used, because I wanted to keep the comparisons based on one consistent set of Original financial-year data and avoid mixing differently processed ABS series in the reported results. As a result, the later-starting Investor Number series did not affect the reported results.

Table 2: Series start and end dates in the original housing finance workbooks

Loan type	Value type	First FY	Last FY	Missing values	Available values
Investor	Number	2019-20	2024-25	1620	648
Investor	Value	2004-05	2024-25	0	2268
Owner-occupier	Number	2004-05	2024-25	0	2268
Owner-occupier	Value	2004-05	2024-25	0	2268

For the Wage Price Index data, the master dataset includes the index series as well as percentage-change series. However, the main analysis in this report uses only the index series, because annual wage growth was calculated consistently from the financial-year average index. This means that the percentage-change series were not used in the reported results.

2.3 Checking Distribution Shapes Across the Datasets

Figure 1 shows no clear sign of major data errors after the asinh transformation. The housing finance and wage price index data have a similar pattern, with slight right-skewness but otherwise fairly smooth distributions and no obvious outliers. In contrast, the overseas migration data are much more concentrated around zero, with longer tails on both sides, which suggests that many observations are small or zero while some are much larger in either the positive or negative direction. Overall, this check suggests that the datasets mainly differ in their distributions rather than showing clear data-quality problems.

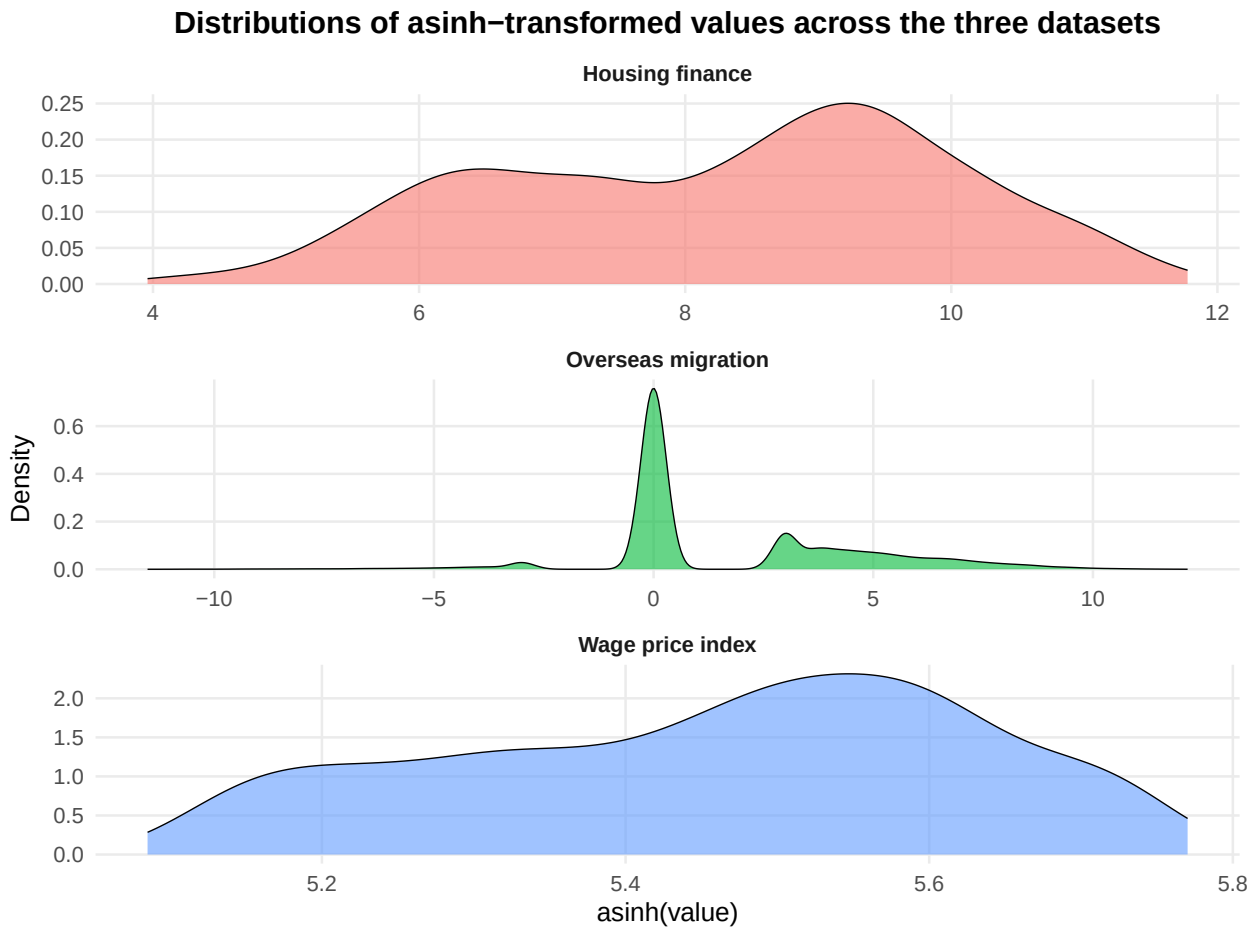

Figure 1

2.4 Summary of Data Checking

Overall, the data wrangling and checking steps suggest that the three datasets were successfully converted into tidy formats and aligned to a common financial-year structure for comparison. No major data quality problems were identified. After completing these checks, I filtered the master datasets to keep only the series used in the main exploration, such as the Original Value series for housing finance and the index series for the Wage Price Index. These cleaned and selected data therefore provide a suitable basis for exploring relationships between migration, housing lending, and wage growth across Australian states.

3 Data Exploration

This section explores the **relationships between housing lending, overseas migration, and wage growth across Australian states** using financial-year summaries and composition measures

from **2004-05 to 2024-25**.

3.1 Housing Lending, Wage Growth and Lending Structure Across Australian States

To address the first research question, I used R to calculate and compare annual housing lending growth and annual wage growth across Australian states from 2005-06 to 2024-25. I then used line plots to examine whether these two measures moved together over time, and to show how the investor share of total lending value changed across states. Further calculation details for these measures are provided in Section A.2.

3.1.1 Divergence between housing lending growth and wage growth

Figure 2 separates housing lending growth and wage growth into two panels, making their state-level patterns easier to compare. Wage growth stays within a relatively narrow range of about 1.5% to 4.2%, but still shows a longer-run decline from 2005-06 to 2020-21 before rising again towards its period average of 3%. Western Australia shows the largest swings in wage growth, while Tasmania appears least affected during the COVID-19 period and rebounds strongly afterwards.

By contrast, housing lending growth is much more volatile and differs more across states. The sharp divergence in 2020-21 likely reflects the COVID-19 period, while the large fall in 2022-23 is consistent with a post-pandemic adjustment. Lending fluctuations are especially large in Northern Territory, Western Australia, and New South Wales, whereas South Australia and the Australian Capital Territory are relatively more stable. Overall, the figure suggests that short-run changes in housing finance are driven much more by lending market conditions than by wage growth.

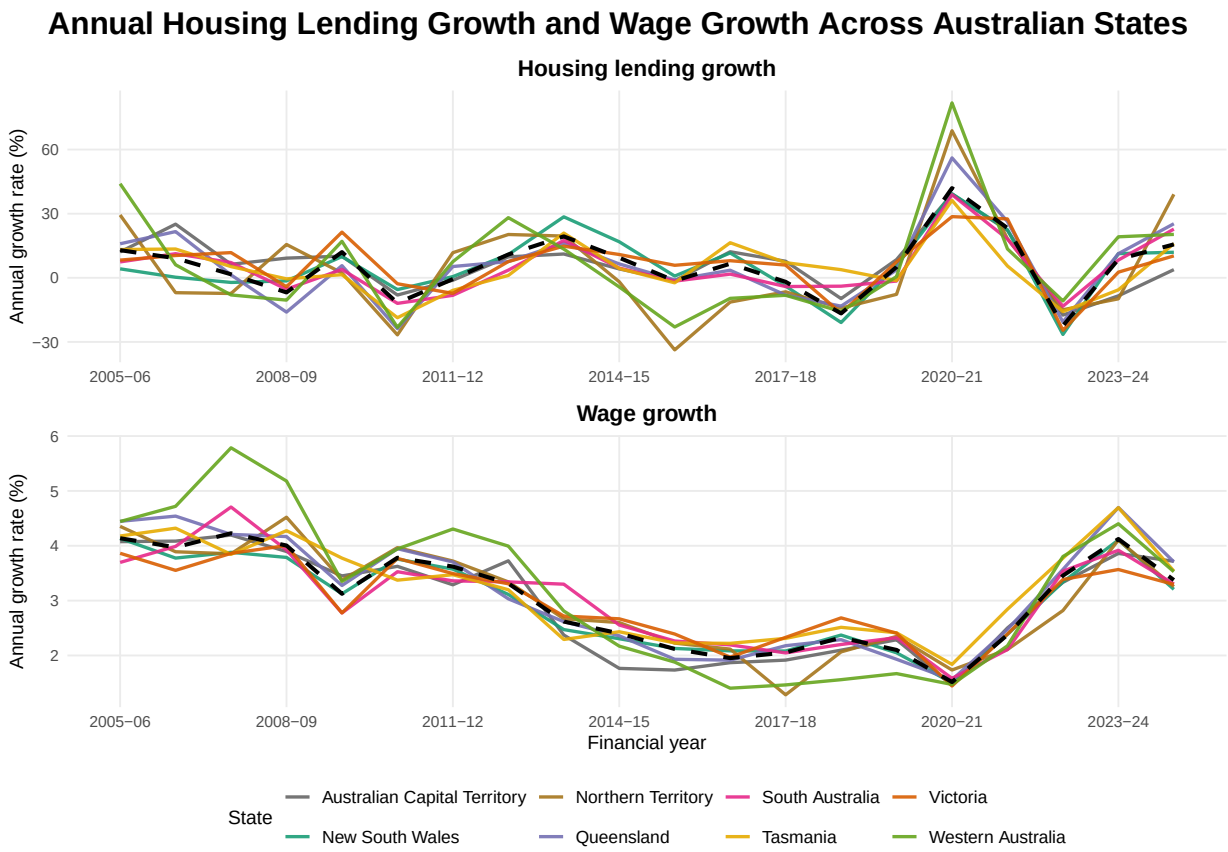

Figure 2: Annual housing lending growth and wage growth across Australian states, 2005-06 to 2024-25. Coloured lines show individual states and the dashed black line shows the Total Australia series.

3.1.2 Long-run changes in the investor share of lending value

Figure 3 examines how the investor share of total housing lending value changed across states from 2004-05 to 2024-25. The national pattern is clearly cyclical. The investor share starts at about 38.3%, rises to a peak of 44.5% in 2014-15, then falls sharply to 25.4% in 2020-21 before recovering to 37.8% by 2024-25.

New South Wales stands out as the most investor-oriented market over the long run. Its average investor share is about 39.6%, and it remains above the national share in 81% of years, peaking at 53.9% in 2014-15. By contrast, Tasmania has one of the lowest average investor shares, at about 25.8%. Northern Territory and Tasmania show a somewhat similar long-run pattern, with investor shares generally remaining below the national benchmark for much of the period. However, both rise sharply after the COVID-19 period. By 2024-25, Northern Territory reaches 40.5% and Tasmania reaches 26.2%, compared with the national share of 37.8%. Western Australia follows a different pattern: after 2008-09, its investor share begins to move further away from the national benchmark, before recovering again after the 2020-21 low. It rises from 16.1% in 2020-21 to 36.8% in 2024-25. The other states generally follow the national cycle more closely and do not show such distinctive departures. Overall, the figure suggests that the structure of housing lending has changed substantially over time and that investor activity has been much more important in some state housing markets than in others.

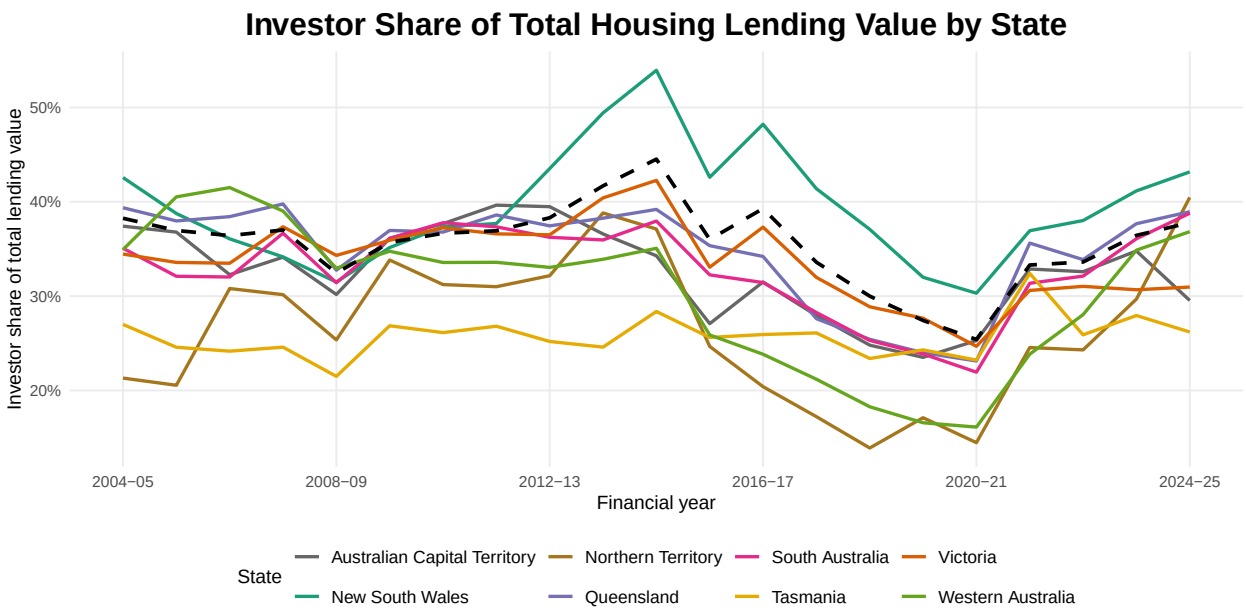

Figure 3: Investor share of total housing lending value across Australian states, 2004-05 to 2024-25. Coloured lines show individual states and the dashed black line shows the Total Australia share.

3.1.3 Summary of findings

Overall, Figure 2 and Figure 3 suggest that housing lending and wage growth followed different state-level dynamics. Lending growth was much more volatile than wage growth, while investor activity shaped the lending structure more strongly in some states than in others, especially New South Wales. Together, these figures suggest that state housing markets differed not only in lending volatility, but also in how strongly investor demand influenced overall lending patterns.

3.2 How have the origins of overseas migrants and their state settlement patterns changed across Australian states over time, and what are their links with wage growth and housing lending?

To address the second research question, I also used R to examined how the origins of overseas migrants changed across Australian states from 2004-05 to 2024-25. I then examined whether migrant-origin concentration within each state was associated with wage growth and housing lending growth.

3.2.1 Changing Origins of Migrant Inflows Across Australian States

The heatmap in Figure 4 shows a clear long-run shift in the origins of migrant inflows across most states. The most notable pattern is the rising importance of Southern and Central Asia, whose national share of positive overseas migrant inflows increases from about 15.6% in 2004-05 to 36.3% in 2024-25. This increase appears across nearly all states, but is especially strong in the Australian Capital Territory and Victoria, where Southern and Central Asia accounts for about 55.6% and 41.0% of positive inflows in 2024-25. By contrast, North-West Europe becomes less important in most states, although it remains more visible in Western Australia, where it still accounts for about 21.0% of positive inflows in 2024-25. Overall, the heatmap suggests that migrant settlement patterns remain uneven across states, with Victoria and the Australian Capital Territory becoming more concentrated in Southern and Central Asian inflows, while Western Australia retains a stronger North-West European component.

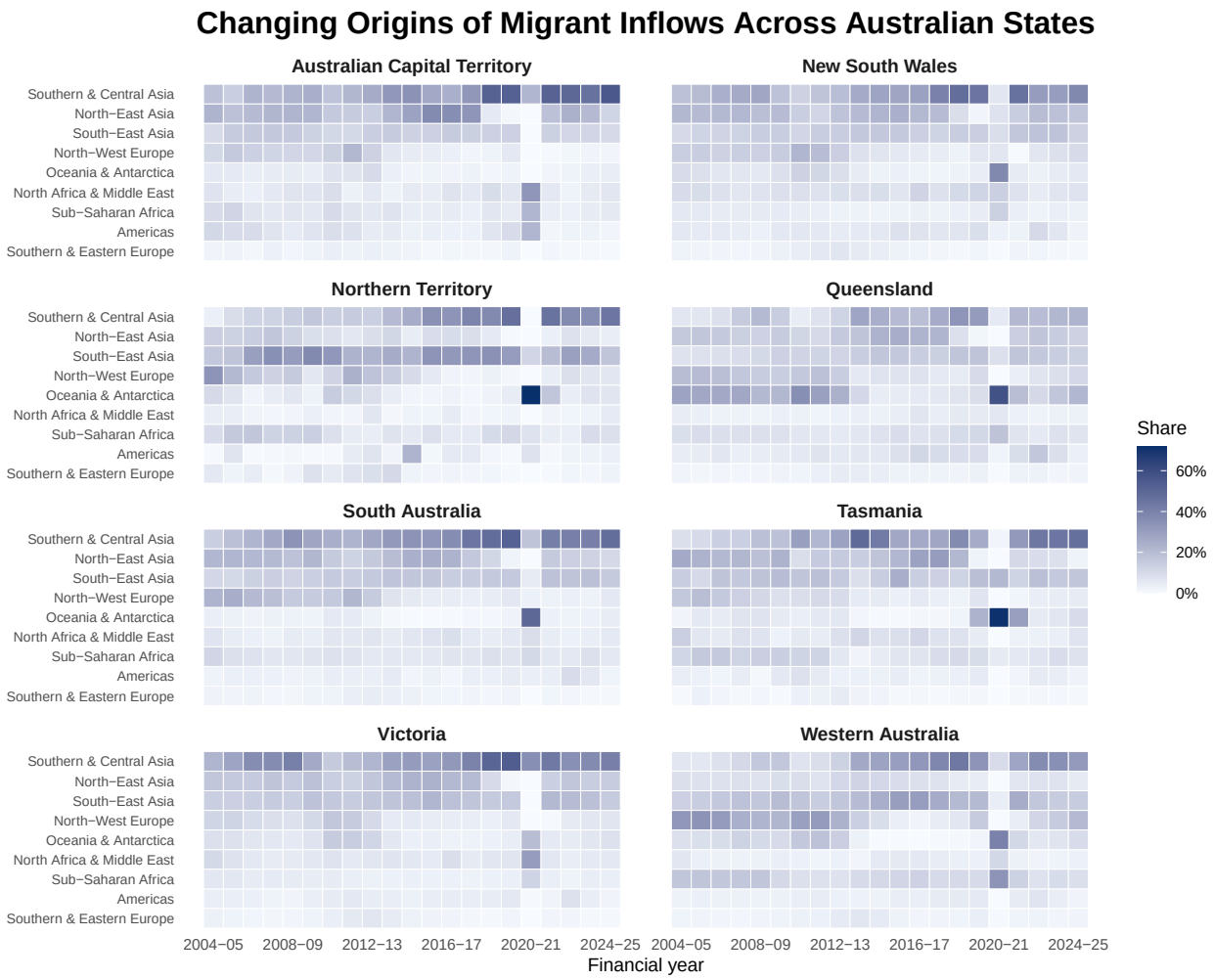

Figure 4: Origin composition of positive overseas migrant inflows across Australian states, 2004-05 to 2024-25. Darker tiles indicate larger shares within each state-year.

3.2.2 Within-state relationships between migrant-origin concentration and economic outcomes

To make the relationship easier to compare, I estimated a separate within-state slope for each state rather than plotting all state-years directly. Migrant-origin concentration was measured using a Herfindahl index (HHI), where higher values indicate that migrant inflows were concentrated in fewer countries of birth. I then examined whether years with relatively higher concentration within each state also tended to have relatively higher wage growth or housing lending growth. To reduce the influence of pandemic-related disruption, I excluded 2020-21 to 2022-23 from this analysis and estimated robust regression models on the remaining years. The results are shown as a coefficient plot with 95% confidence intervals in Figure 5.

Further calculation details for this within-state analysis are provided in Section A.3, and additional diagnostics for the excluded years are reported in Section A.1.

In Figure 5, positive coefficients indicate that, within a state, years with more concentrated migrant inflows tended to coincide with higher economic outcomes, while negative coefficients indicate the opposite. In the wage-growth panel, most states still have negative coefficients, which suggests that more concentrated migrant inflows were often linked to lower wage growth within states. This pattern is most visible in New South Wales, Tasmania, the Australian Capital Territory, and Northern Territory, where the intervals stay below zero. Queensland stands out as the main exception, showing a positive relationship, while the other states are closer to zero and less clear.

The housing-lending panel is more mixed. Most states are on the negative side of zero, but the intervals are wide and usually cross zero, so the relationship is not very clear in most cases. Tasmania is the main exception, because it remains clearly positive and stands out from the other states. Overall, this figure suggests that migrant-origin concentration has a clearer relationship with wage growth than with housing lending growth, but the pattern still differs across states.

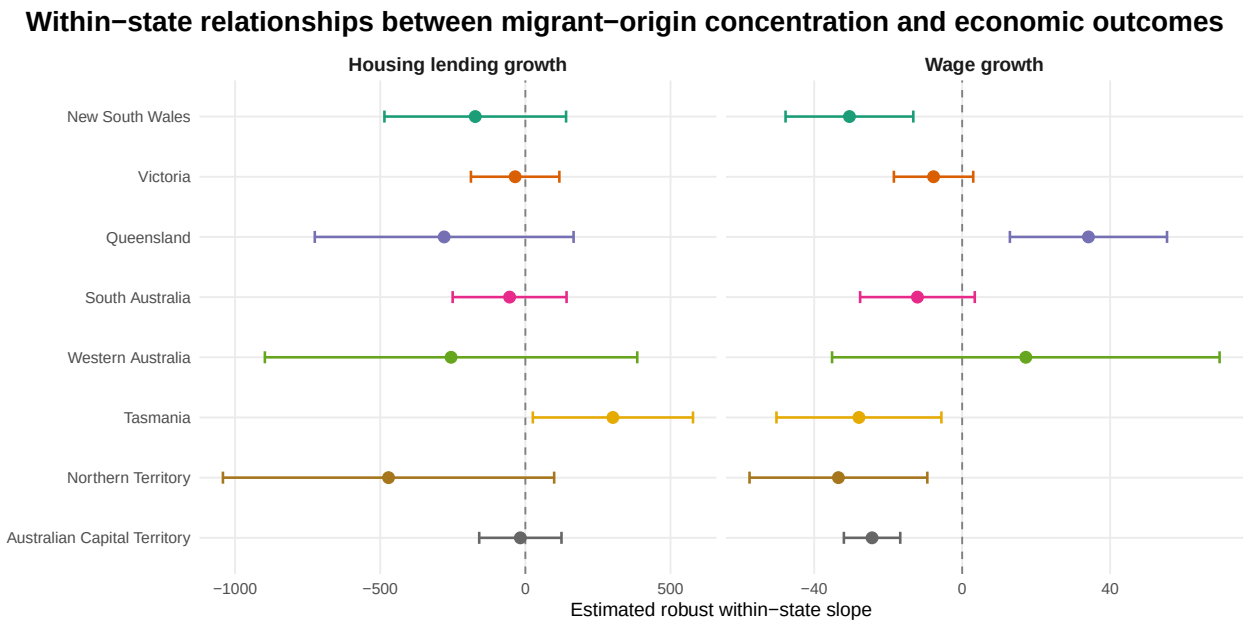

Figure 5: Robust within-state associations between migrant-origin concentration and economic outcomes, excluding 2020-21 to 2022-23. Coefficients are estimated from state-specific robust regressions using deviations from each state's own average level. Positive coefficients indicate that years with above-average migrant-origin concentration tend to coincide with above-average economic growth within the same state.

3.2.3 Summary of findings

Overall, Figure 4 and Figure 5 suggest that migrant origins and settlement patterns changed clearly over time, but their economic links were less uniform across states. Southern and Central Asia became more prominent in most states, especially in Victoria and the Australian Capital Territory, while Western Australia retained a more visible North-West European component. The within-state results suggest that migrant-origin concentration was more clearly related to wage growth than to housing lending growth, but these relationships still varied across states. Overall, changes in migrant composition were important, but their economic effects were uneven rather than consistent across Australia.

3.3 Where are the largest gaps between investor lending and migration demand across Australian states, and how has this geography changed over time?

To reduce the influence of unusually disrupted migration patterns during the pandemic and the immediate reopening period, I excluded 2020-21 to 2022-23 from the mapped comparison. In this section, migration demand is proxied by each state's share of positive overseas migrant inflows, and the gap is defined as investor lending share minus migration-demand share. Using R, I calculated these measures for each state and year, then averaged them within three representative periods: an earlier period (2004-05 to 2011-12), a middle period (2012-13 to 2019-20), and a recent period (2023-24 to 2024-25). I then exported the summarised state-level table to Tableau and visualised the results as a set of choropleth maps using a diverging colour scheme centred at zero. Averaging within each period makes the differences across states easier to compare visually.

Further calculation details for this gap measure are provided in Section A.4, and the excluded years are summarised separately in Section A.1.

In Figure 6, blue states indicate that investor lending is stronger than migration demand, while orange states indicate the opposite. The largest positive gap in the early period is in Queensland, while the largest negative gap is in Western Australia. In the middle and recent periods, New South Wales has the largest positive gap, whereas Victoria has the largest negative gap. This shows that the geography of mismatch becomes more concentrated over time. Queensland remains positive throughout, although the gap becomes smaller in the middle period before returning to a stronger positive value in the recent period. Western Australia remains negative across all three periods. Northern Territory stays close to neutral overall but is slightly negative in the middle and recent periods. The Australian Capital Territory remains close to neutral but slightly positive throughout, while Tasmania shifts from slightly positive in the early period to slightly negative in the later periods. South Australia is less stable, moving from slightly positive to negative in the middle period before returning to near neutral in the recent period.

Overall, Figure 6 suggests that the investor-migration gap became less even over time. The strongest mismatches are concentrated in a small number of states, especially New South Wales and Victoria, but in opposite directions. This pattern is consistent with Figure 3, where New South Wales appears as the most investor-oriented market, and it also complements Figure 4, which shows that migrant settlement patterns remain uneven across states. However, the supplementary statistical test provides weaker support for this result, so it should be interpreted more cautiously. Even so, the figure still suggests that investor activity and migration-related housing demand were not distributed evenly across Australian states, which helps explain why housing pressure may be experienced differently across state housing markets.

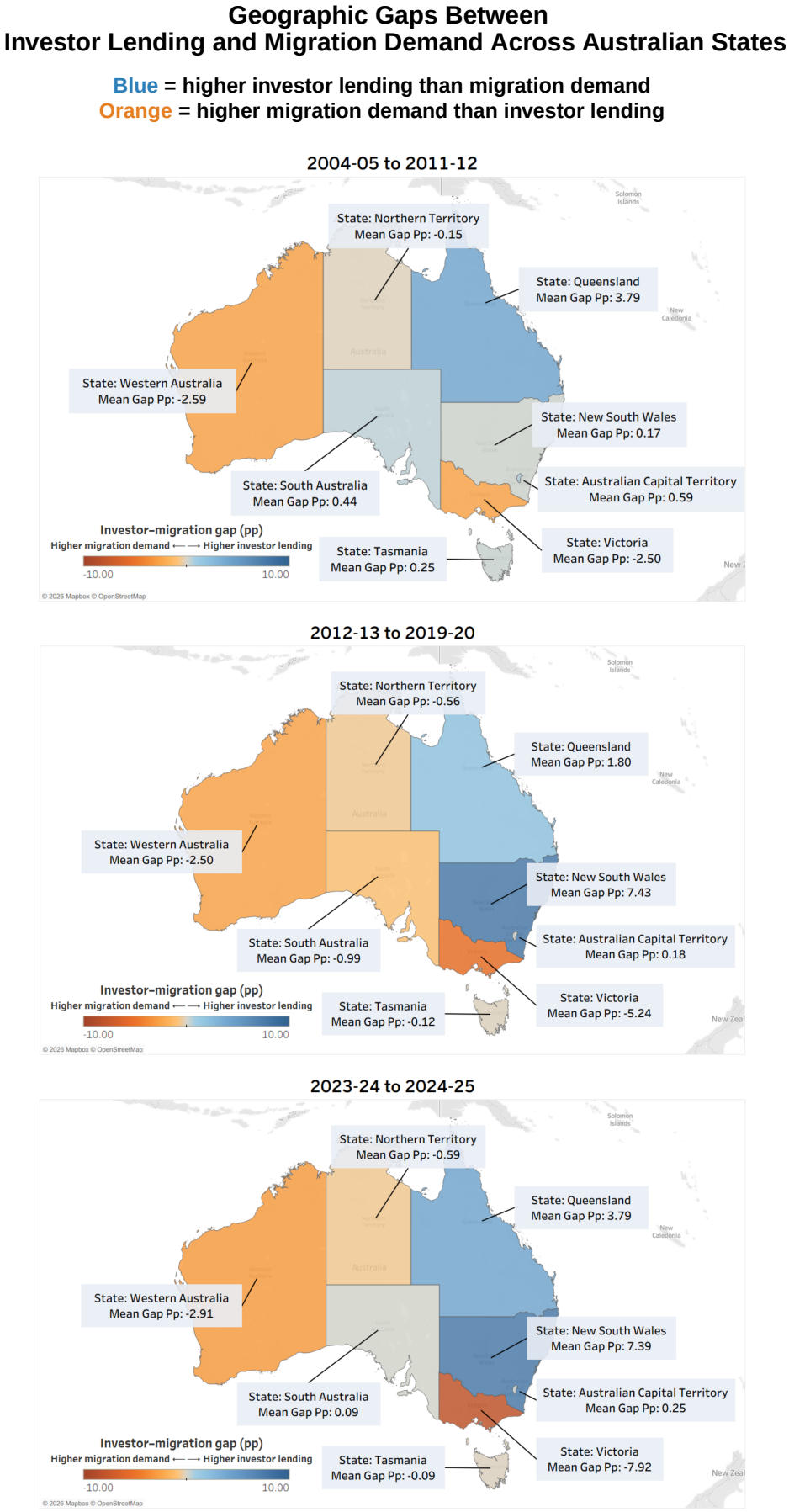

Figure 6: Average gaps between investor lending share and migration-demand share across Australian states in three representative periods. Positive values indicate that a state's share of investor lending exceeded its share of migration demand, while negative values indicate the opposite.

4 Conclusion

This report shows that housing lending, overseas migration and wage growth did not move together in one simple national pattern across Australian states. The exploration showed that housing lending growth was much more volatile than wage growth (Figure 2), while the investor share of lending changed markedly over time and remained especially important in some states, particularly New South Wales (Figure 3). Migrant origins also changed clearly over the study period, with Southern and Central Asia becoming more prominent across many states (Figure 4), but the within-state links between migrant-origin concentration and economic outcomes were uneven and were generally clearer for wage growth than for housing lending growth (Figure 5).

Taken together with Figure 6, these findings show that the gap between investor lending and migration demand became more geographically concentrated over time, especially in New South Wales and Victoria. Overall, the original questions suggest that housing lending does not closely align with wage growth, migrant settlement patterns have shifted substantially, and the mismatch between investor activity and migration demand differs strongly across states. In this report, *Housing Lending, Overseas Migration and Wage Growth Across Australian States* is best understood as a state-specific rather than uniform national story. These conclusions are also broadly consistent with the supplementary statistical checks in Section A.5, although the additional support is weaker for the Question 3 result. These results are descriptive rather than causal, but they show that the relationships at the centre of Australia's housing debate vary substantially across state housing markets.

5 Reflection

This project was challenging in many ways, but I also learned a great deal from it. The most difficult part was the page limit. As a Master of Business Analytics student, I use R frequently and have some background in statistics, so I usually like exploring data from multiple angles. However, in this assignment I had to answer three large research questions within a very limited space. This forced me to think much more carefully about what to include and what to leave out. Through this process, I realised that data exploration is not just about doing as much analysis as possible, but also about selecting the clearest and most meaningful way to communicate the main story.

Another challenge was deciding how to calculate and summarise the data so that each question could be answered well with only one or two figures. This taught me an important lesson about data visualisation. A good visualisation is not just about making more plots, but about choosing the right calculation and the most suitable visual form. In this project, I learned to use calculations more carefully so that one figure could answer more than one part of a question, which helped me reduce duplicate or repetitive plots and make the report more concise. I also improved a lot in debugging, because the ABS datasets were complicated to clean and combine, and many parts of the analysis required repeated calculations and custom functions. If I did this project again, I would plan the structure earlier, decide on the final visualisations sooner, and think earlier about how each figure could answer the research question as directly as possible.

6 References

Australian Bureau of Statistics. (n.d.). Australian Bureau of Statistics. Retrieved April 15, 2026, from https://www.abs.gov.au/

Szumer, Z. (2026, March 3). 'Pushed out of paradise': High migration is often blamed for the housing crisis. Is that right? SBS News. https://www.sbs.com.au/news/insight/article/is-high-migration-really-causing-the-housing-crisis/4i6dsl8yh

7 Package citations

Garrett Grolemund, Hadley Wickham (2011). Dates and Times Made Easy with lubridate. Journal of Statistical Software, 40(3), 1-25. https://www.jstatsoft.org/v40/i03/.

Ooms J (2026). magick: Advanced Graphics and Image-Processing in R. doi:10.32614/CRAN.package.magick https://doi.org/10.32614/CRAN.package.magick, R package version 2.9.1, https://CRAN.R-project.org/package=magick.

Pedersen T (2025). patchwork: The Composer of Plots. doi:10.32614/CRAN.package.patchwork https://doi.org/10.32614/CRAN.package.patchwork, R package version 1.3.2, https://CRAN.R-project.org/package=patchwork.

R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Venables, W. N. & Ripley, B. D. (2002) Modern Applied Statistics with S. Fourth Edition. Springer, New York. ISBN 0-387-95457-0

Wickham H (2023). conflicted: An Alternative Conflict Resolution Strategy. doi:10.32614/CRAN.package.conflicted https://doi.org/10.32614/CRAN.package.conflicted, R package version 1.2.0, https://CRAN.R-project.org/package=conflicted.

Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E, Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the tidyverse." Journal of Open Source Software, 4(43), 1686. doi:10.21105/joss.01686 https://doi.org/10.21105/joss.01686.

Wickham H, Bryan J (2025). readxl: Read Excel Files. doi:10.32614/CRAN.package.readxl https://doi.org/10.32614/CRAN.package.readxl, R package version 1.4.5, https://CRAN.R-project.org/package=readxl.

Wickham H, Pedersen T, Seidel D (2025). scales: Scale Functions for Visualization. doi:10.32614/CRAN.package.scales https://doi.org/10.32614/CRAN.package.scales , R package version 1.4.0, https://CRAN.R-project.org/package=scales.

Wilke C (2025). cowplot: Streamlined Plot Theme and Plot Annotations for 'ggplot2'. doi:10.32614/CRAN.package.cowplot https://doi.org/10.32614/CRAN.package.cowplot, R package version 1.2.0, https://CRAN.R-project.org/package=cowplot.

Wilke C, Wiernik B (2022). gridtext: Improved Text Rendering Support for 'Grid' Graphics. doi:10.32614/CRAN.package.gridtext https://doi.org/10.32614/CRAN.package.gridtext, R package version 0.1.5, https://CRAN.R-project.org/package=gridtext.

Xie Y (2025). knitr: A General-Purpose Package for Dynamic Report Generation in R. R package version 1.51, https://yihui.org/knitr/.

Zhu H (2024). kableExtra: Construct Complex Table with 'kable' and Pipe Syntax. doi:10.32614/CRAN.package.kableExtra https://doi.org/10.32614/CRAN.package.kableExtra, R

package version 1.4.0, https://CRAN.R-project.org/package=kableExtra.

8 AI statement

I mainly used Positron while writing my code, so I relied on its built-in Codex assistant for debugging, coding support, grammar checking, code revision, and some LaTeX writing. I also used Gemini mainly when reading in data, especially for writing functions, and for a few questions that Codex did not answer well.

In my experience, Codex was a very useful support tool. It also helped me improve the presentation of some plots and make them look more polished. Gemini was especially helpful for writing functions, as its code was often more organised and concise than Codex's. At the same time, I did not rely too heavily on these AI tools, because they sometimes generated unnecessary code or over-processed the analysis. In some cases, this made the plots look more complicated and the code harder to debug. Because of this, I still needed to evaluate their suggestions carefully and decide what was actually useful for my project.

Below are some sample prompts I used during the project.

Functions

- Help me turn this time series dataset from wide to long format and join it to the metadata table. I have already added year, month, and financial_year from the date column before pivoting the data into long format, but I also need to add a loan_type argument to the function. Can you help me revise the function?
- I am reading multiple ABS Excel files with similar but not identical structures. Can you help me write a reusable function that reshapes the data from wide to long format, joins it to the metadata, and includes a loan_type argument?
- Please help me simplify this function for reading and tidying time series data. I want it to create year, month, and financial_year, pivot the data to long format, join the metadata table, and keep the code concise.

Documentation / LaTeX / Grammar

- Please help me write these equations in proper LaTeX so they can be included in a PDF report.
- Please help me fix the grammar of this paragraph and make it sound more academic but still concise.

Coding / Debugging

- Why am I getting this error in R? Can you explain the cause and show me how to fix it?
- I got this error when running my code. Please help me identify what is wrong and suggest a fix.
- This code is not working as expected. Can you help me debug it and explain the problem step by step?

Revise charts / figure layout

- I have a map saved as a PDF file called figure/comb_map.pdf in R. I want to display this PDF figure, add a title above it that says "Geographic Gaps Between Investor Lending and Migration Demand Across Australian States", and combine them into one final plot. Can you help me do this?
- I have a PDF map in R and I want to add a title above it, then combine both into a single final figure for my report. Can you show me how to do this using magick and cowplot?

- I already created a figure by combining a PDF map and a title in R, but the title is too high. How can I move the title lower?
- Can you help me improve the layout of this figure?
- I want the title and subtitle to sit closer to the figure and look more balanced.

A Appendix

A.1 Diagnostics and excluded-year checks for Questions 2 and 3

This section reports additional checks for the observations excluded from Questions 2 and 3. For Question 2, I report influence diagnostics for 2020-21. For Question 3, I report the investor-migration gaps for 2020-21 directly.

A.1.1 Q2: Influence diagnostics for 2020-21

For Question 2, I estimated state-specific linear models on the full sample and calculated leverage and Cook's distance for each state-year observation. These diagnostics show whether 2020-21 had an unusually strong influence on the fitted slopes. The final coefficients in Figure 5 were still estimated using robust regression after excluding 2020-21 to 2022-23.

Table 3: Leverage and Cook's distance for 2020-21 in the Question 2 within-state analysis

State	Outcome	FY	Lev	Lev cut	Cook	Cook cut	High lev	High Cook
Northern Territory	Housing lending growth	2020-21	0.976	0.2	70.418	0.2	Yes	Yes
Western Australia	Housing lending growth	2020-21	0.483	0.2	2.999	0.2	Yes	Yes
Queensland	Housing lending growth	2020-21	0.248	0.2	1.300	0.2	Yes	Yes
Australian Capital Territory	Housing lending growth	2020-21	0.226	0.2	0.904	0.2	Yes	Yes
Tasmania	Housing lending growth	2020-21	0.787	0.2	0.760	0.2	Yes	Yes
Victoria	Housing lending growth	2020-21	0.312	0.2	0.650	0.2	Yes	Yes
South Australia	Housing lending growth	2020-21	0.075	0.2	0.299	0.2	No	Yes
New South Wales	Housing lending growth	2020-21	0.062	0.2	0.156	0.2	No	No
Northern Territory	Wage growth	2020-21	0.976	0.2	96.098	0.2	Yes	Yes
Tasmania	Wage growth	2020-21	0.787	0.2	3.553	0.2	Yes	Yes
Western Australia	Wage growth	2020-21	0.483	0.2	1.054	0.2	Yes	Yes
Queensland	Wage growth	2020-21	0.248	0.2	0.839	0.2	Yes	Yes
Victoria	Wage growth	2020-21	0.312	0.2	0.457	0.2	Yes	Yes
New South Wales	Wage growth	2020-21	0.062	0.2	0.208	0.2	No	Yes
South Australia	Wage growth	2020-21	0.075	0.2	0.105	0.2	No	No
Australian Capital Territory	Wage growth	2020-21	0.226	0.2	0.000	0.2	Yes	No

A.1.2 Q3: Investor-migration gaps in 2020-21

For Question 3, Table 4 reports the investor-migration gap for 2020-21 directly. This helps show why the pandemic year was treated as atypical and excluded from the mapped comparison periods.

Table 4: Investor–migration gaps in the excluded year for Question 3

State	Financial year	Investor share (%)	Migration share (%)	Gap (pp)
Australian Capital Territory	2020-21	2.4	0.4	2.0
New South Wales	2020-21	42.2	62.8	-20.6
Northern Territory	2020-21	0.3	1.1	-0.8
Queensland	2020-21	16.8	15.1	1.7
South Australia	2020-21	4.5	5.4	-0.9
Tasmania	2020-21	1.3	2.5	-1.2
Victoria	2020-21	26.7	6.1	20.6

Western Australia	2020-21	5.8	6.5	-0.7

A.2 Further calculation details for Housing Lending, Wage Growth and Lending Structure

For Question 1, annual wage growth was calculated from the financial-year average of the Wage Price Index:

$$WageGrowth_{st} = 100 \times \left(\frac{WPI_{st}}{WPI_{s,t-1}} - 1 \right)$$

Annual housing lending growth was calculated from the total annual value of housing lending in each state:

$$LendingGrowth_{st} = 100 \times \left(\frac{Lending_{st}}{Lending_{s,t-1}} - 1 \right)$$

The investor share of lending value was calculated as:

$$InvestorShare_{st} = \frac{InvestorValue_{st}}{InvestorValue_{st} + OwnerOccupierValue_{st}}$$

These measures were based on the Original Value series and were used to compare lending growth, wage growth, and lending structure across states over time.

A.3 Further calculation details for Within-State Links Between Migrant-Origin Concentration and Economic Outcomes

For each state and financial year, migrant-origin concentration was measured using a country-level Herfindahl index:

$$HHI_{st} = \sum_i p_{ist}^2$$

where p_{ist} is the share of positive net overseas migration from country of birth i in state s and year t:

$$p_{ist} = \frac{\max(\text{NOM}_{ist}, 0)}{\sum_j \max(\text{NOM}_{jst}, 0)}$$

The within-state analysis excludes 2020-21 to 2022-23. After excluding these years, I centred migrant-origin concentration, wage growth, and housing lending growth around each state's own average:

$$HHI_{st}^{dev} = HHI_{st} - \bar{HHI}_s$$

$$WAGE_{st}^{dev} = WAGE_{st} - \overline{WAGE}_s$$

$$LEND_{st}^{dev} = LEND_{st} - \overline{LEND}_s$$

I then estimated separate robust regression models for each state:

$$WAGE_{st}^{dev} = \alpha_s + \beta_s HHI_{st}^{dev} + \varepsilon_{st}$$

$$LEND_{st}^{dev} = \alpha_s + \beta_s HHI_{st}^{dev} + \varepsilon_{st}$$

The coefficient β_s shows whether years with above-average migrant-origin concentration within a state also tended to have above-average wage growth or housing lending growth. The intervals shown in the figure are approximate 95% confidence intervals.

A.4 Further calculation details for Geographic Gaps Between Investor Lending and Migration Demand

For each state s and financial year t, migration demand was proxied by the state share of positive overseas migrant inflows:

$$MigrationShare_{st} = \frac{PositiveNOM_{st}}{\sum_r PositiveNOM_{rt}}$$

Investor activity was measured by the state share of investor lending value:

$$InvestorShare_{st} = \frac{InvestorValue_{st}}{\sum_r InvestorValue_{rt}}$$

The investor-migration gap was then calculated as:

$$Gap_{st} = 100 \times (InvestorShare_{st} - MigrationShare_{st})$$

The mapped comparison excludes 2020-21 to 2022-23 and summarises three representative periods: 2004-05 to 2011-12, 2012-13 to 2019-20, and 2023-24 to 2024-25. For each state and period, the mapped value is the average annual gap:

$$\overline{Gap}_{sp} = \frac{1}{T_p} \sum_{t \in p} Gap_{st}$$

A.5 Supplementary statistical checks

This section reports supplementary statistical checks for the main conclusions in the report. These tests are used as robustness checks rather than formal proof. Because the project is exploratory and based on full ABS datasets, some conclusions are better treated as descriptive summaries than as hypotheses to be formally tested.

A.5.1 Q1: Permutation tests for the alignment between housing lending growth and wage growth

To check whether housing lending growth and wage growth moved closely together within states, I first matched the two annual growth series by state and financial year. For each state, I calculated the observed Pearson correlation between housing lending growth and wage growth. I then created a permutation distribution by keeping the wage-growth series fixed and randomly shuffling the housing-lending-growth series 5,000 times, recalculating the correlation each time. The permutation p-value is the proportion of shuffled correlations that were at least as extreme as the observed correlation. Large p-values suggest that the observed relationship could easily arise by chance, which is consistent with the conclusion that the two series did not closely align.

Table 5: Permutation tests for within-state correlations between housing lending growth and wage growth

State	Observed correlation	Permutation p-value
Australian Capital Territory	-0.190	0.419
New South Wales	-0.376	0.098
Northern Territory	-0.004	0.988
Queensland	-0.179	0.450
South Australia	-0.206	0.385
Tasmania	-0.381	0.098
Victoria	-0.441	0.053
Western Australia	-0.012	0.960

To support the conclusion that New South Wales was more investor-oriented than the national benchmark, I compared its investor share with the national investor share in each financial year. I then counted the number of years in which New South Wales was above the national benchmark. Under a simple null benchmark of 50%, New South Wales would be expected to be above the national share in about half of all years. I used an exact binomial test to assess whether the observed proportion was substantially greater than this benchmark. A small p-value supports the interpretation that New South Wales was above the national investor share more often than would be expected by chance.

Table 6: Exact binomial test for New South Wales exceeding the national investor share

years_above_national	total_years	proportion_above	p_value
17	21	0.81	0.0036

A.5.2 Q2: Permutation test for the rise of Southern and Central Asia in national migrant inflows

To support the conclusion that Southern and Central Asia became more prominent over time, I used the national yearly share of positive inflows from this region and compared it with the order of financial years. I first calculated the observed Spearman correlation between the time index and the yearly share, since this statistic is suitable for checking a monotonic trend over time. I then randomly shuffled the time index 5,000 times and recalculated the Spearman correlation each time to form a permutation distribution. The p-value is the proportion of shuffled correlations that were at least as large as the observed one. A strong positive association with a small p-value supports the long-run upward shift seen in the heatmap.

Table 7: Permutation test for the upward trend in the Southern and Central Asia share of national positive migrant inflows

observed_spearman	p_value
0.639	0.0012

A.5.3 Q2: Permutation tests for within-state slopes

To check whether the within-state relationships between migrant-origin concentration and economic outcomes were stronger than expected by chance, I used the state-level centred data from the main

analysis. For each state and each outcome, I first estimated the observed slope from a simple linear regression of the centred outcome on centred migrant-origin concentration. I then kept the HHI series fixed and randomly shuffled the outcome values 5,000 times, recalculating the slope each time. This produced a permutation distribution under the assumption that there was no systematic relationship between the two variables within that state. The permutation p-value is the proportion of shuffled slopes that were at least as extreme as the observed slope. Smaller p-values indicate stronger evidence that the observed within-state relationship was unlikely to arise by chance.

Table 8: Permutation tests for within-state slopes between migrant-origin concentration and economic outcomes

State	Outcome	Observed slope	Permutation p-value
Australian Capital Territory	Wage growth	-24.13	0.000
New South Wales	Wage growth	-30.47	0.003
Northern Territory	Wage growth	-33.56	0.009
Queensland	Wage growth	28.12	0.039
South Australia	Wage growth	-11.30	0.186
Tasmania	Wage growth	-27.91	0.031
Victoria	Wage growth	-7.26	0.178
Western Australia	Wage growth	17.45	0.482
Australian Capital Territory	Housing lending growth	-22.74	0.749
New South Wales	Housing lending growth	-235.32	0.144
Northern Territory	Housing lending growth	-473.00	0.098
Queensland	Housing lending growth	-246.16	0.188
South Australia	Housing lending growth	-32.94	0.738
Tasmania	Housing lending growth	328.94	0.031
Victoria	Housing lending growth	-68.11	0.387
Western Australia	Housing lending growth	-0.39	0.998

A.5.4 Q3: Randomisation test for whether the investor-migration gap became less even over time

To support the conclusion that the geography of the investor-migration gap became less even over time, I compared the cross-state dispersion of the average gap in the early period and the recent period. I first reshaped the state-level averages so that each state had one early-period value and one recent-period value, and then calculated the observed difference in variance between the two periods. To assess whether this increase could arise by chance, I used an exact randomisation test that swaps the early and recent labels within each state in all possible combinations. For each relabelling, I recalculated the variance difference to generate the reference distribution. The p-value is the proportion of randomised variance differences that were at least as large as the observed one. A positive observed difference with a small p-value supports the conclusion that the gap became more geographically concentrated over time.

Table 9: Exact randomisation test for whether the cross-state dispersion of the investor-migration gap was larger in the recent period than in the early period

Observed Variance Difference	P value
16.094	0.0781